

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1596

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I Vishali.s(192521049) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *vishali*

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks-2

Experiments

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

AIM:

To create a simple cloud-based Hospital Information System for SIMATS Hospital using a Cloud Service Provider to demonstrate Software as a Service (SaaS) with patient and hospital management details.

Procedure

- Log in to the cloud platform (Zoho Creator).
- Create a new application named **SIMATS Hospital Information System**.
- Create a patient registration form.
- Add the required fields:
 - Patient Name
 - Age
 - Address
 - Phone Number
 - Email
 - Attender Name
 - Attender Phone
 - Doctor Name
 - Disease
- Save the form and publish the application.
- Enter sample patient details and verify that the data is stored successfully in the cloud database.
- Test the application by adding, editing, and viewing patient records.

1.1 Patient details

HMSHOSPITAL
MANAGMENT...

Patient Details

Patient Details

Patient Details Report

Doctor Details

Appointment Details

Patient Details Report

	Patient ID	Patient Name	Age	Gender	Blood Group	Phone Number	Address
	4	sanjay	17	Male	O+	+91384757594939	circular block, chennai, tamil nadu, 612098, India
	3	dharshini	19	Female	AB+	+9110294857432	saveetha nagar ,school backside, thandalam, tamil nadu, 612304, Briti Ocean Territory
	2	jefna	22	Female	B-	+911929384763	st john street, kadalur, tamil nadu, 632019, India
	1	vishali	18	Female	O-	+911219983737	198f kamarajar street vallar nagae , vellore, tamil nadu, 632602, India

1.2 doctor details

HMSHOSPITAL
MANAGMENT...

Patient Details

Doctor Details

Doctor Details

Doctor Details Report

Appointment Details

Doctor Details Report

	Doctor ID	Doctor Name	Department	Qualification	Exprience	Mobile number	Email ID
	4	poorna	Orthpedics	bbbs	13	+91345678912	poorna@gmail.com
	3	ramesh	Dermatology	mbbs,dbbs	3	+91234567891	ramesh@gamil.com
	2	suresh dinesh	ENT	mbbs,psd	12	+91987654321	suresh@gmail.com
	1	honey ini	Orthpedics	mbbs	23	+91123456789	hpney@gmail.com

1.3 appointment details

HMSHOSPITAL
MANAGMENT...

Patient Details

Doctor Details

Appointment Details

Appointment Details Report

Appointment Details Report

	Appointment ID	Patient Details	Doctor Details	Appointment Date	Appointment Time	Disease	Status
	3	dharshini	suresh dinesh	05-Aug-2026	16:36:40	lungs problem	Booked
	2	vishali	honey ini		07:31:00	heart attack	Completed
	1	sanjay	ramesh	30-Jul-2026	10:35:31	fever	Booked

- **Result**

A cloud-based Hospital Information System was successfully created using a Cloud Service Provider to demonstrate the SaaS model. The application allows users to store and manage hospital and patient information online.

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

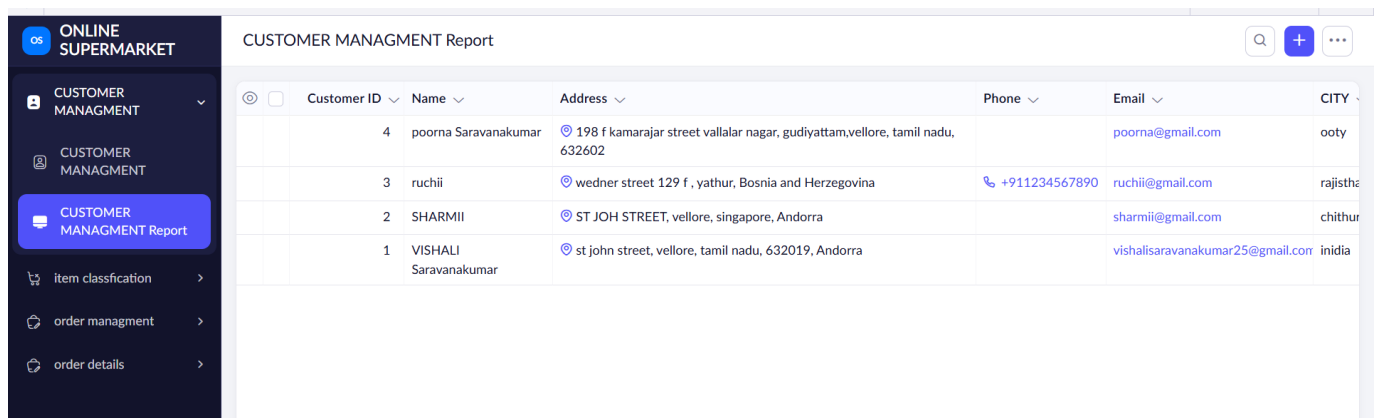
Aim

To create a simple cloud-based Online Supermarket application using a Cloud Service Provider to demonstrate Software as a Service (SaaS) with customer details, item details, quantity, and item classification.

Procedure

- Log in to the cloud platform (Zoho Creator).
- Create a new application named **Online Supermarket System**.
- Create a customer order form.
- Add the required fields:
- Customer Name
- Address
- Phone Number
- Email
- Item Name
- Quantity
- Item Classification
- Save the application and publish it.
- Enter sample customer and product details.
- Verify that the order details are stored and can be viewed or edited through the cloud application.

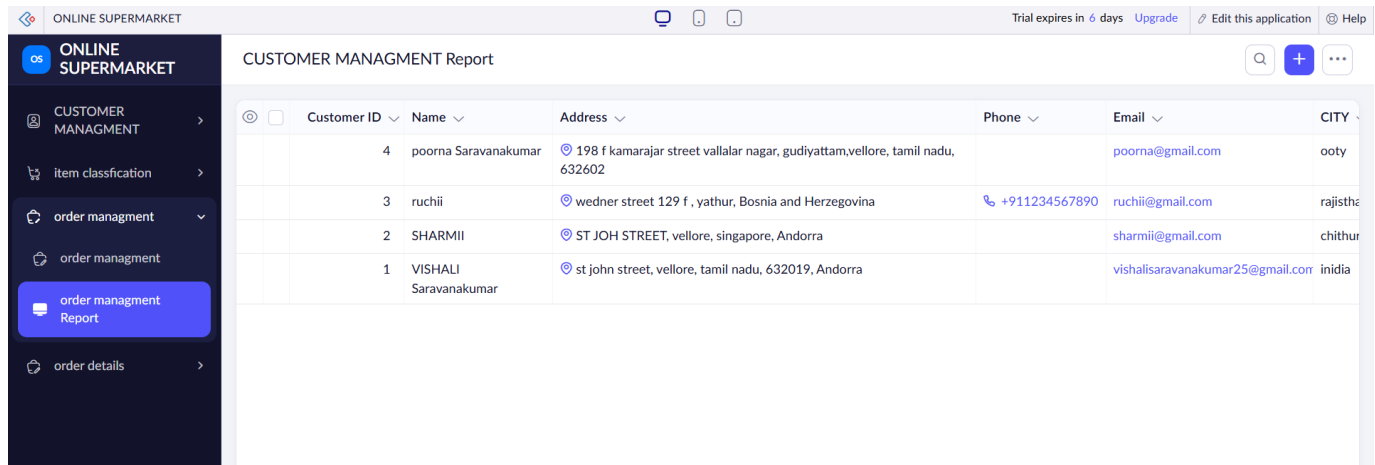
2.1 customer management



The screenshot displays the 'ONLINE SUPERMARKET' application interface. On the left is a dark sidebar with navigation options: 'CUSTOMER MANAGEMENT', 'CUSTOMER MANAGEMENT Report' (highlighted), 'item classification', 'order management', and 'order details'. The main area shows a 'CUSTOMER MANAGEMENT Report' table with columns for Customer ID, Name, Address, Phone, Email, and CITY. The table contains four rows of customer data.

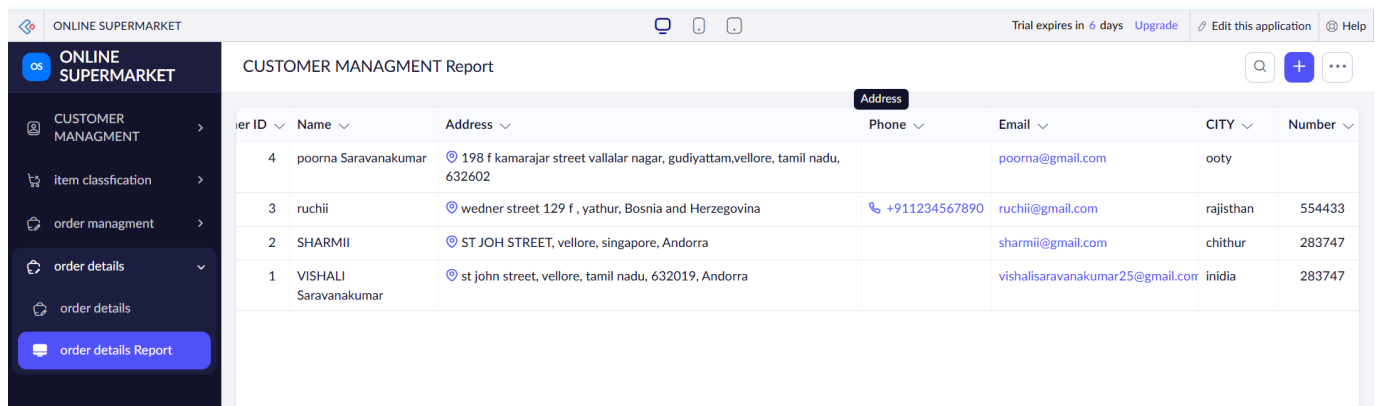
Customer ID	Name	Address	Phone	Email	CITY
4	poorna Saravanakumar	198 f kamarajar street vallalar nagar, gudiyattam,vellore, tamil nadu, 632602		poorna@gmail.com	ooty
3	ruchii	wedner street 129 f., yathur, Bosnia and Herzegovina	+911234567890	ruchii@gmail.com	rajistha
2	SHARMII	ST JOH STREET, vellore, singapore, Andorra		sharmii@gmail.com	chithur
1	VISHALI Saravanakumar	st john street, vellore, tamil nadu, 632019, Andorra		vishalisaravanakumar25@gmail.com	inidia

2.2 customer management



Customer ID	Name	Address	Phone	Email	CITY
4	poorna Saravanakumar	198 f kamarajar street vallalar nagar, gudiyattam,vellore, tamil nadu, 632602		poorna@gmail.com	ooty
3	ruchii	wedner street 129 f , yathur, Bosnia and Herzegovina	+911234567890	ruchii@gmail.com	rajistha
2	SHARMII	ST JOH STREET, vellore, singapore, Andorra		sharmii@gmail.com	chithur
1	VISHALI Saravanakumar	st john street, vellore, tamil nadu, 632019, Andorra		vishalisaravanakumar25@gmail.com	inidia

2.3 customer management



Customer ID	Name	Address	Phone	Email	CITY	Number
4	poorna Saravanakumar	198 f kamarajar street vallalar nagar, gudiyattam,vellore, tamil nadu, 632602		poorna@gmail.com	ooty	
3	ruchii	wedner street 129 f , yathur, Bosnia and Herzegovina	+911234567890	ruchii@gmail.com	rajistha	554433
2	SHARMII	ST JOH STREET, vellore, singapore, Andorra		sharmii@gmail.com	chithur	283747
1	VISHALI Saravanakumar	st john street, vellore, tamil nadu, 632019, Andorra		vishalisaravanakumar25@gmail.com	inidia	283747

Result

A cloud-based Online Supermarket application was successfully developed using a Cloud Service Provider to demonstrate the SaaS model. The application enables online storage and management of customer and product information.

These write-ups are concise and suitable for a practical record, matching the style typically expected in cloud computing laboratory experiments.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

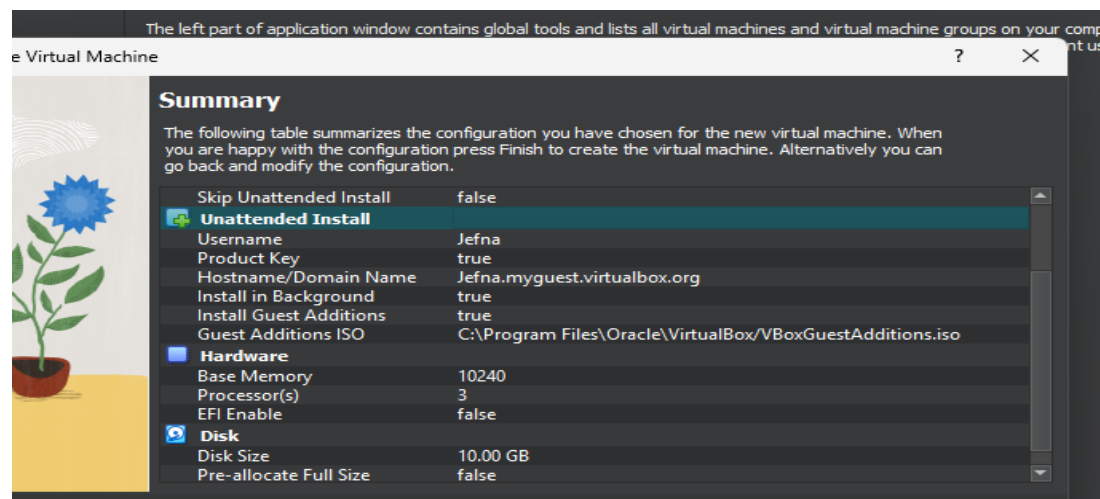
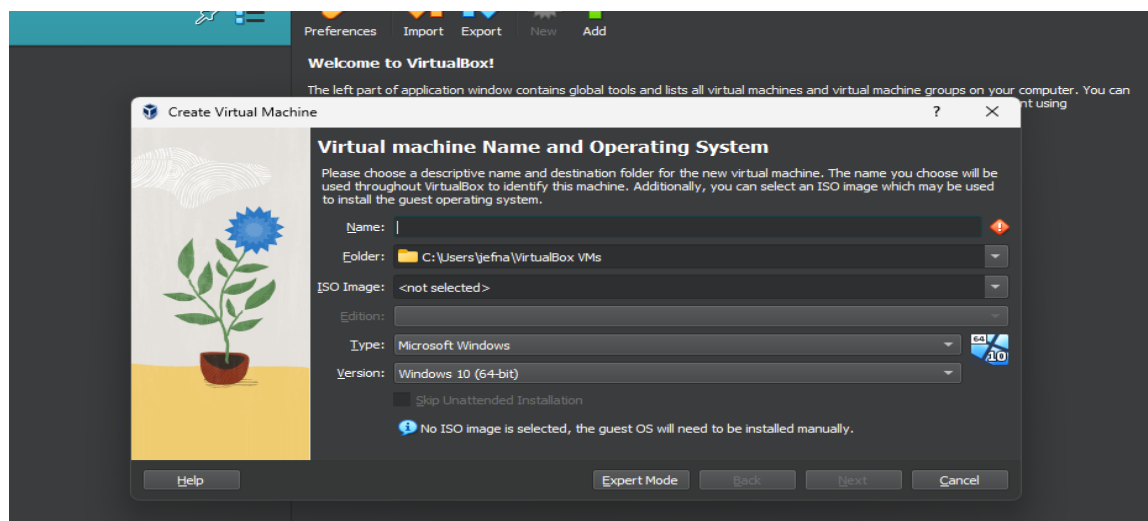
Aim

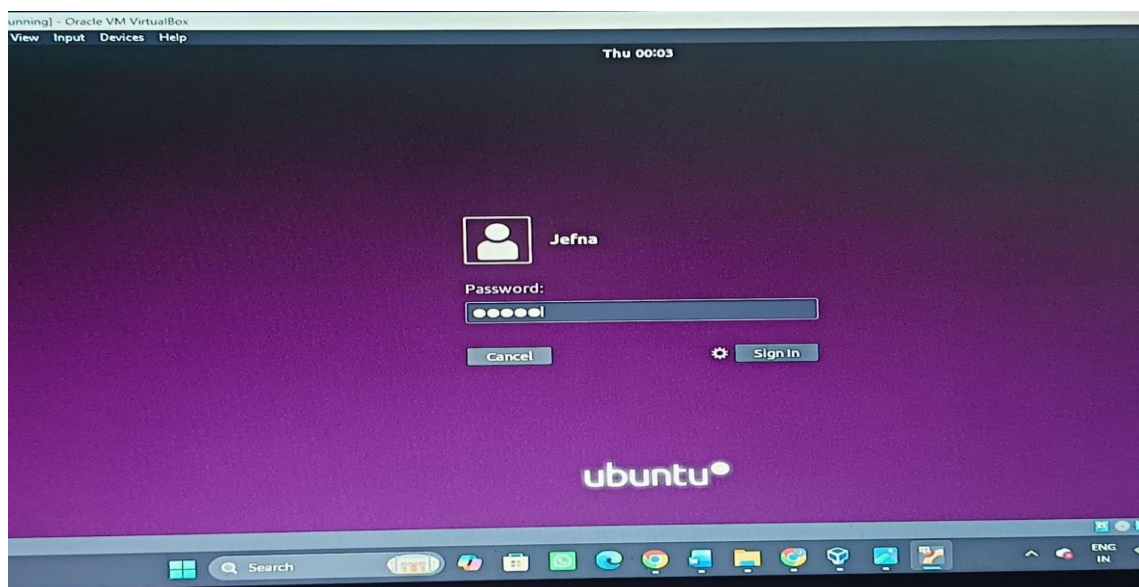
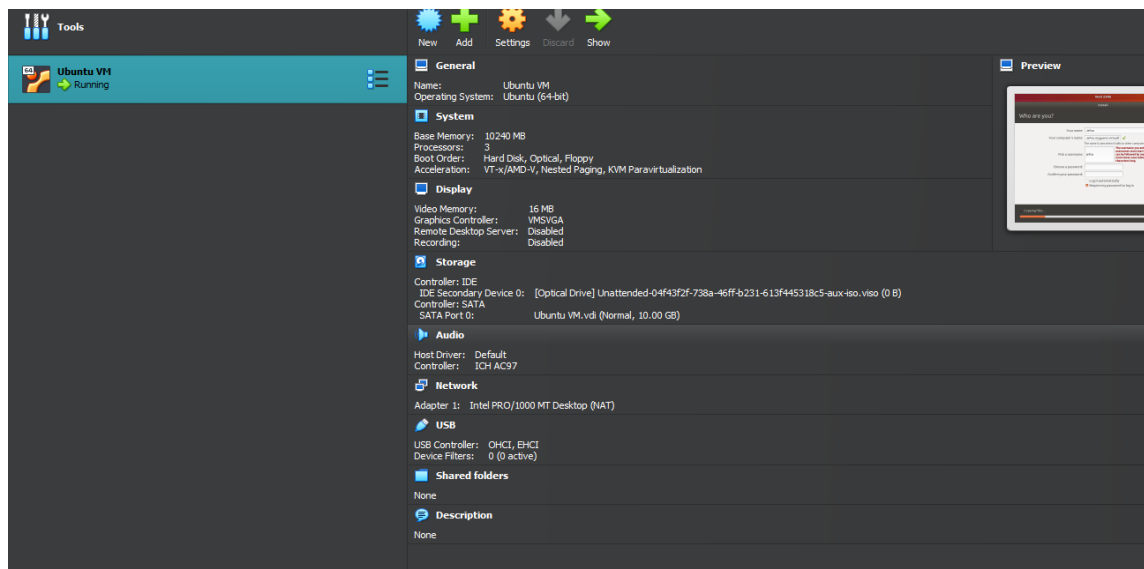
To demonstrate virtualization by installing a Type-2 Hypervisor, creating and configuring a virtual machine with 3 CPU cores, 10 GB RAM, and 10 GB virtual storage using Oracle VirtualBox.

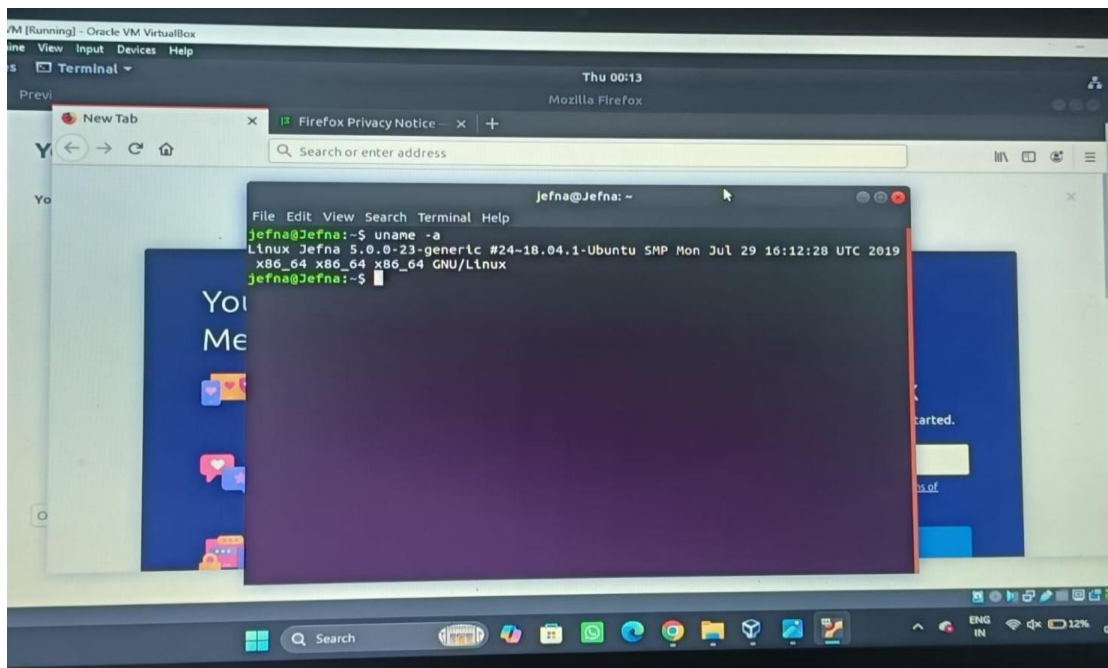
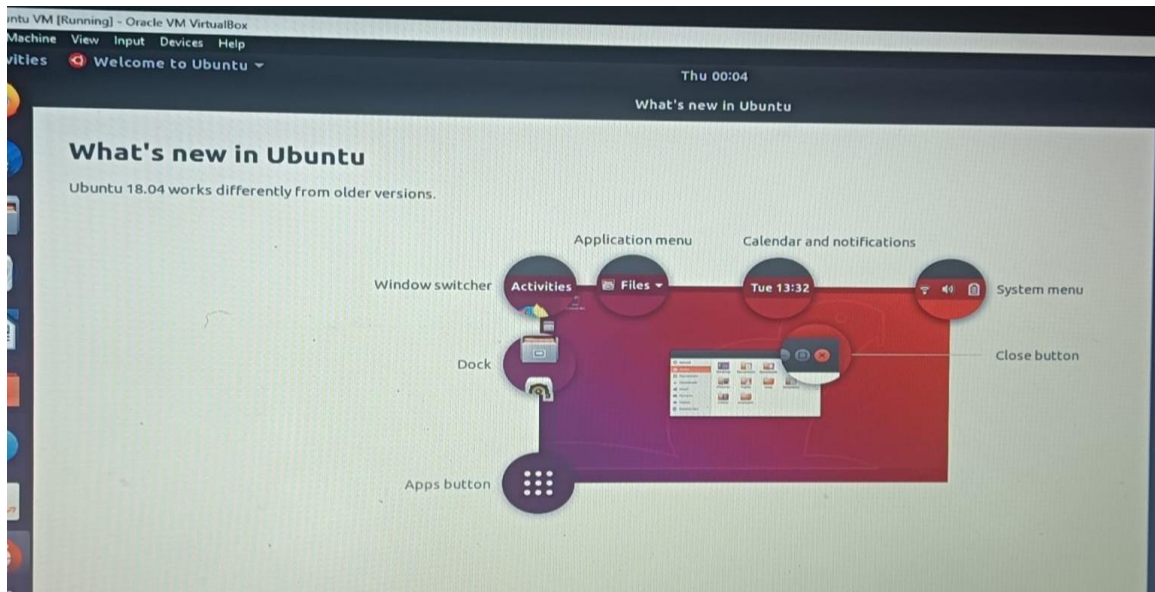
Procedure

1. Download and install Oracle VirtualBox on the host operating system.
2. Download the Ubuntu Linux ISO image.
3. Open VirtualBox and create a new virtual machine.
4. Select Linux as the operating system.
5. Allocate 10 GB (10240 MB) RAM.
6. Assign 3 CPU cores to the virtual machine.
7. Create a 10 GB virtual hard disk.
8. Attach the Ubuntu ISO file to the virtual machine.
9. Start the virtual machine and install Ubuntu Linux.
10. Verify that the virtual machine boots successfully.

Output:







Result

Thus, Type-2 virtualization was successfully demonstrated by installing Oracle VirtualBox on the host operating system. A virtual machine was created and configured with 3 CPU cores, 10 GB RAM, and a 10 GB virtual hard disk, and the Ubuntu Linux operating system was installed successfully. Hence, the virtual machine was created and executed successfully.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

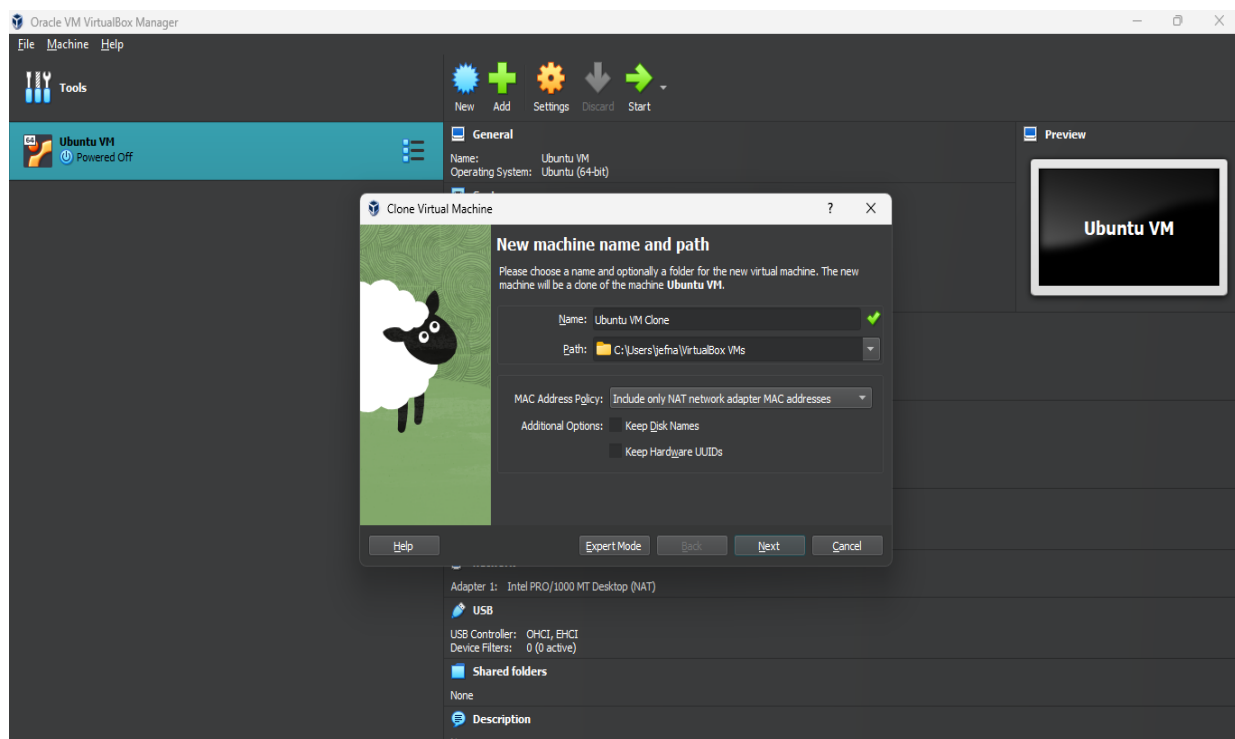
Aim

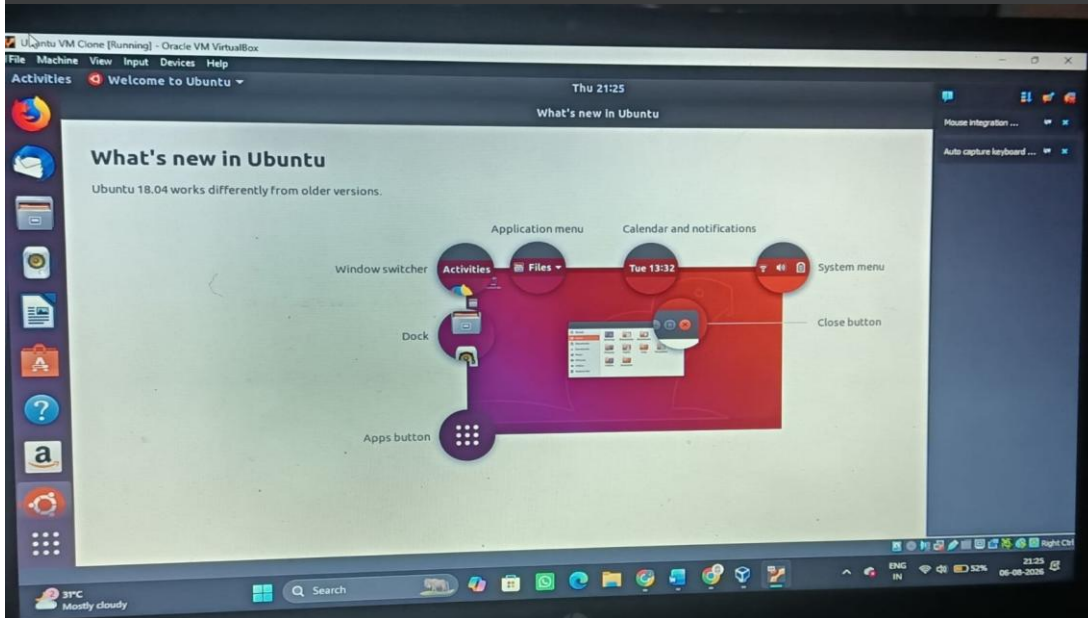
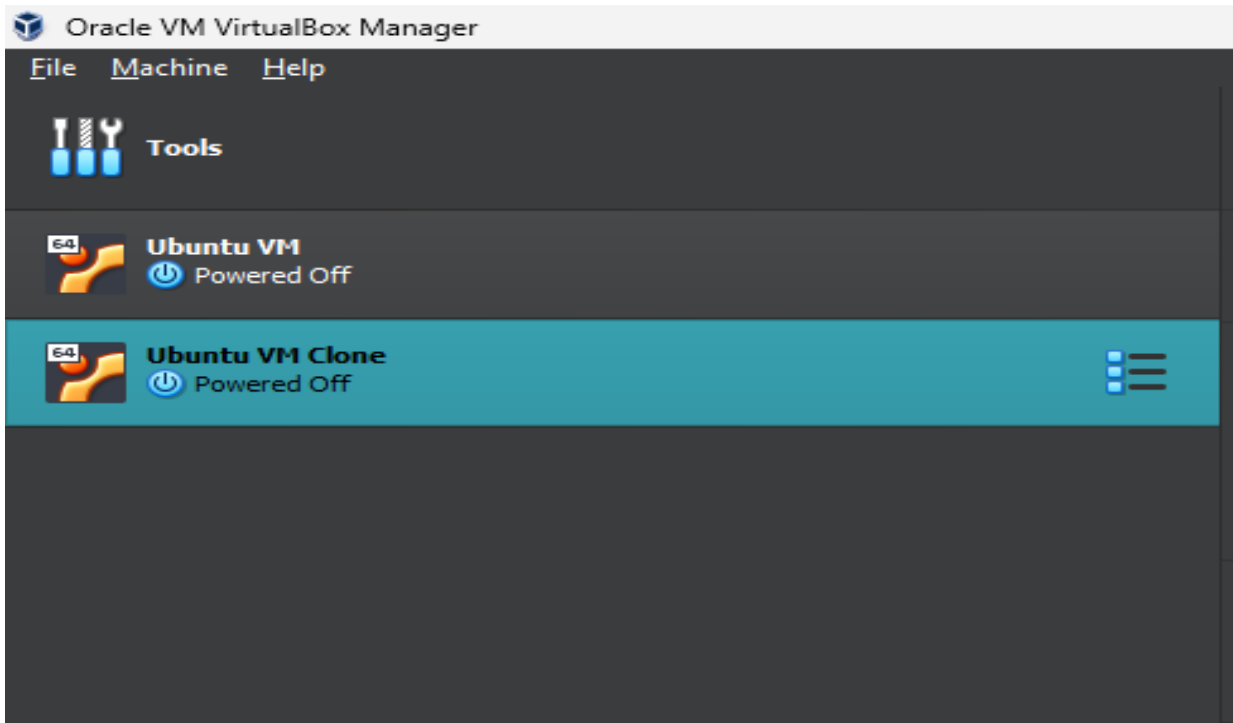
Clone your existing Ubuntu Virtual Machine and prove that the cloned VM works.

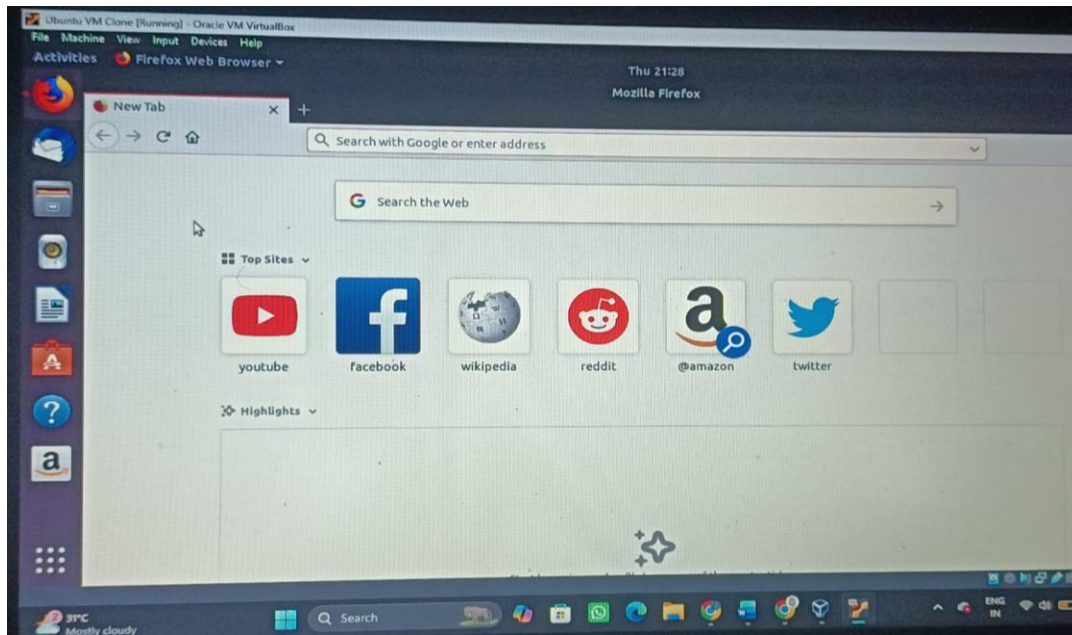
Procedure

1. Open Oracle VM VirtualBox on the host operating system.
2. Ensure the existing virtual machine (Ubuntu VM) is in the Powered Off state.
3. Select the Ubuntu VM from the left panel.
4. Click Machine → Clone (or right-click the VM and select Clone).
5. Enter a name for the cloned virtual machine (e.g., Ubuntu Clone).
6. Click Next and choose Full Clone as the cloning type.
7. Keep the default MAC Address Policy and click Finish.
8. Wait until the cloning process is completed successfully.
9. Verify that the cloned virtual machine appears in the VirtualBox Manager.
10. Select the Ubuntu Clone and click Start.
11. Wait for the cloned VM to boot and verify that the operating system loads successfully.
12. Shut down the cloned virtual machine after testing.

Output:







Result

The virtual machine was successfully cloned using Oracle VM VirtualBox, and the cloned VM was tested successfully by booting into the Ubuntu operating system.